



Cornell University

Principles of Biochemistry: Proteins and Metabolism

BioMG 3310/6311, 3 credits, Letter or S/U

Fall, 2026

Class Meetings: 10:10 – 11:00 am Monday, Wednesday and Friday, Uris Hall G01

Instructor: **Professor Linda Nicholson**, Office location: **210 Stimson Hall**, E-mail: **LKN2@cornell.edu**
Professor Nicholson's Office Hours (a welcoming and inclusive space): **1:30 – 3:30 Mondays**.

Meets Distribution Requirements: (BSC-AG, OPHLS-AG), (BIO-AS)

Prerequisites: one majors-level biology course and one year general chemistry and any of the following organic chemistry courses: CHEM 1570 or CHEM 3530 or CHEM 3570 or CHEM 3590 or equivalent, or permission of instructor.

Forbidden Overlap: students who receive credit for the BIOMG 3310/3320 sequence will not receive credit for any of the following courses: BIOMG 3300, BIOMG 3350, NS 3200.

Course Description

In this lecture-based course, students start from fundamental principles of chemistry, physics, and mathematics as the framework for understanding biology. Lectures and activities will enable students to assimilate the fundamental principles while they systematically master the complexity of biochemistry. Emphasis is placed on the interconnections of vast amounts of information, particularly in metabolism, that are the basis for life.

Learning Outcomes (BioMG 3310/6311)

After completing the course, students will be able to:

- Draw key biomolecules (amino acids, lipids, carbohydrates).
- Predict the metabolic fate of fuel molecules under given conditions.
- Apply quantitative principles to solve problems in protein stability, molecular transport, redox reactions, and enzyme kinetics.
- Analyze protein and carbohydrate structures downloaded from the Protein Data Bank using ChimeraX and interpret the results in terms of biochemical principles.

Additional Learning Outcomes for BioMG 6311

After completing the course, graduate students enrolled in 6311 additionally will be able to:

- Critique primary source material on a relevant biochemical topic.
- Convey their detailed critique in the form of a written commentary article.

Teaching Assistants

Graduate TAs (gTAs):

NAME	Weekly Review Time/Place	Office Hour time, place	EMAIL
Isaac Lamptey	Tuesdays, 3:30 – 5:00/TBA	Tuesdays, 5:00 – 7:00 PM/Weill Hall 467	inl8
Maddie Ceminsky	Tuesdays, 3:30 – 5:00/TBA	Wednesdays, 2:30 – 3:30 PM/Plant Sci G44	mcc295

Undergraduate TA (uTA) Study Group Leaders:

NAME	Study Group Time/Place	EMAIL
Lamisa Aziz	Wednesdays, 5:00 - 6:30/GSH 181 (w/Sneha)	la464
Jaden Bascon	Sundays, 3:00 - 4:30 PM/GSH 181 (w/Joshua)	jrb526
Emily Bauer	Mondays, 3:30-5:00 PM/TBA (w/Mahealani)	esb284
Noah Bove	Wednesdays, 5:00 - 6:30/Kennedy 103 (w/Noah)	ntb52
Emily Buckley	Wednesdays, 6:30 - 8:00 PM/Malott 205 (w/Ann)	emb432
Giancarlo Chuquitarco	Sundays, 5:00 - 6:30/GSH 181 (w/Lucy)	gc539
Mahealani Demello	Mondays, 3:30-5:00 PM/TBA (w/Emily Bauer)	mld259
Angel Escamilla	Saturdays, 1:00 - 2:30 PM/Malott 230 (w/Jonathan)	ae428
Jonathan Falcon Cruz	Saturdays, 1:00 - 2:30 PM/Malott 230 (w/Angel)	jf874
Landon Fried	Mondays, 6:30 - 8:00 PM/GSH 181 (w/Katherine)	lrf66
Marissa Gaut	Sundays, 1:00 - 2:30PM/GSH 181 (w/Justin)	mlg323
Maya Gowda	Thursdays, 12:30 – 2:00 PM/TBA (w/Aleha)	msg283
Katie Greening	Mondays, 2:45 - 4:15 PM/TBA (w/Aryan)	kmg299
Ann Greenwood	Wednesdays, 6:30 - 8:00 PM/Malott 205 (w/Emily Buckley)	aeg277
Lucy Katarina Hemesath	Sundays, 5:00 - 6:30/GSH 181 (w/Giancarlo)	lkh56
Chloe Hickman	Wednesdays, 2:00 – 3:30 PM/TBA (w/Arzu)	czh5
Jeffrey Jiang	Fridays, 12:30 - 2:00 PM/Stimson 106 (w/Allie)	jj724
Sneha Katikaneni	Wednesdays, 5:00 - 6:30/GSH 181 (w/Lamisa)	ssk264
Katherine Kraus	Mondays, 6:30 - 8:00 PM/GSH 181 (w/Landon)	ksk92
Brian Lee	Fridays, 3:00 - 4:30 PM/Stimson 106	hl989
Wendy Li	Saturdays, 11:00 AM - 12:30 PM/GSH 181 (w/Rena)	wl735
Allie Pequeno	Fridays, 12:30 - 2:00 PM/Stimson 106 (w/Jeffrey)	atp73
Aryan Rajput	Mondays, 2:45 - 4:15 PM/TBA (w/Katie)	agr89
Aleha Syed	Thursdays, 12:30 – 2:00 PM/TBA (w/Maya)	ais78
Justin Taveras	Sundays, 1:00 - 2:30PM/GSH 181 (w/Marissa)	jt874
Max Taylor	Wednesdays, 5:00 - 6:30/Kennedy 103 (w/Noah)	mwt56
Arzu Thareja	Wednesdays, 2:00 – 3:30 PM/TBA (w/Chloe)	aat86
Joshua Tincher	Sundays, 3:00 - 4:30 PM/GSH 181 (w/Jaden)	jwt99
Rena Wang	Saturdays, 11:00 AM - 12:30 PM/GSH 181 (w/Wendy)	rhw229

TBA = To Be Announced (rooms are requested but have not yet been assigned)

Course Schedule (subject to change as needed during the semester!)

Lecture Topics, Assignments, Exams

(reading assignments are posted for each lecture in the weekly modules on Canvas site)

LECTURE #	DATE	TOPIC	LG Pages	LG Problems, Case Studies, ChimeraX
1	M, 8/24	introduction	7-23	
2	W, 8/26	amino acids, ionization, pH, pK _a , "the 20"	24-37	Case Study #1: pH Peril
3	F, 8/28	peptides	38-45	1-3
4	M, 8/31	intro to proteins	46-53	ChimeraX Quiz (Q) 1 (Tutorial) due
5	W, 9/2*	intro to protein structure	54-60	
6	F, 9/4	protein structure	61-76	8, 10-13, ChimeraX Q2 (Sec. Str.) due
NO CLASS	M, 9/7	LABOR DAY	-----	
7	W, 9/9	protein structure	77-86	Clicker Participation Points Begin!
8	F, 9/11	protein structure	87-99	9, 14-16, 18, ChimeraX Q3 (Protein folds) due
9	M, 9/14	thermodynamic stability	100-115	
10	W, 9/16*	protein folding	116-125	Case Study #2: Observation over Intuition
11	F, 9/18	intro to hemoglobin	126-135	4, 19 - 22, ChimeraX Q4 (Hemoglobin) due
12	M, 9/21	hemoglobin	136-147	
13	W, 9/23*	Intro to enzymes	148-161	
14	F, 9/25	principles of enzyme kinetics	162-174	23 - 28, ChimeraX Q5 (Protein Data Bank) due
15	M, 9/28	enzymes and MM kinetics	175-187	
16	W, 9/30	enzyme inhibition	188-194	Case Study #3: Toxic Alcohols
PRELIM 1	R, 10/1	COVERS 8/24-9/25		
17	F, 10/2	chymotrypsin mechanism	195-209	29 - 33
18	M, 10/5	enzymes - cofactors, regulation	210-221	ChimeraX Q6 (Chymotrypsin) due
19	W, 10/7*	zymogens / intro to lipids	222-231	
20	F, 10/9	lipids and biomembranes	232-238	34 - 37
NO CLASS	M, 10/12	FALL BREAK	-----	
Catch Up	W, 10/14	Catch-Up Lecture	up thru 238	
21	F, 10/16	carbohydrates	239-251	38 - 42
DROP deadline	M, 10/19			
22	M, 10/19	intro to bioenergetics	252-261	ChimeraX Q7 (Carbohydrates) due
23	W, 10/21*	bioenergetics, redox rxns	262-272	
24	F, 10/23	glycolysis	273-284	39,43,45
25	M, 10/26	glycolysis	285-293	
26	W, 10/28*	metabolic regulation	294-299	Case Study #4: Sudden Onset
27	F, 10/30	glycolysis reg. - PFK-1 and PK	300-306	44,46, ChimeraX Q8 (PFK-1) due
28	M, 11/2	pyr. dehyd. complex	307-316	
Catch Up	W, 11/4*	Catch-Up Lecture		
PRELIM 2	R, 11/5	COVERS 9/28-10/30		
29	F, 11/6	citric acid cycle	317-326	47a, 48,49, Case Study #5: An Unexplained Death
30	M, 11/9	fat catabolism	327-335	Case Study #6: A Day at the Beach
31	W, 11/11	amino acid ox; urea cycle	336-344	
32	F, 11/13	ox. phosphorylation	345-356	47b,50-54
33	M, 11/16	ox. phosphorylation	357-369	Case Study #7: The Narrow Window
34	W, 11/18*	photophosphorylation	370-377	
35	F, 11/20	photophosphorylation	378-385	55 - 59, ChimeraX Q9 (ATP Synthase) due
36	M, 11/23	carbohydrate biosynth.	386-398	60 - 61
NO CLASS	W, 11/25	THANKSGIVING BREAK	-----	
NO CLASS	F, 11/27	THANKSGIVING BREAK	-----	
37	M, 11/30	lipid biosynth.	399-411	
38	W, 12/2*	biosyn. of N-containing mol.	412-418	
39	F, 12/4	regulation of metabolism	419-430	62 - 65, Case Study #8 (Run. Exp), ChiX Q10 (G-Prot.) due
40	M, 12/7	COURSE SUMMARY/Catch-Up	431-435	3 Case Studies DUE! (top 3 scores kept)
FINAL	TBA	FULLY CUMULATIVE		

***IN CLASS QUIZ DAY (10 total, top 6 scores count, Wed. Quiz Keys posted Fri., NO MAKEUPS)**

*Bibliographic and Other Course Materials (**Required**):*

- Text: **Nelson & Cox, Lehninger Principles of Biochemistry, 8th Edition**. YOU **MUST either get the eBook with Achieve via CAMP**, or if you **opted out of CAMP**, you **MUST purchase it [here](#)** (you need the **\$124.99** option that includes Achieve learning materials AND eBook, otherwise it won't work for the course).
- A **physical iClicker**. Note: **Cornell's CAMP includes one iClicker**.
- **Lecture Guide**: PDFs (one complete and separate smaller sections) are posted on the BIOMG 3310 Canvas site.

Inclusivity Statement:

This course strives to create an inclusive and collaborative learning environment that promotes civil, respectful, and supportive interactions within and outside of our classroom.

Course Support and Supplemental Instruction:

- Graduate TAs will hold a weekly Review Session and office hours (TBA). Material from the prior week will be reviewed during the 90-minute in-person sessions.
- Undergraduate TAs will lead 90-minute Study Groups (*sign up for one!*) where they will cover the key concepts of the prior week and guide collaborative learning and problem solving.
- Graduate students are encouraged to make use of the Knight Writing Center as they work on their Commentary article.

Expectations:

- Use of Generative Artificial Intelligence (GAI):
 - OK for enhancing your leaning/understanding of concepts **while studying**
 - WARNING: GAI is not always accurate!
 - DO NOT use GAI to generate answers to ANY OF your assignments --- this “shortcut” will not prepare you for in-class assessments where you have no access to GAI.
- Group/collaborative work:
 - respect each other; be kind; be generous; challenge the idea, not the person
 - use your active listening skills, make sure all voices in the group are heard
 - tap into the advantages of experiencing different perspectives
- Attend Class! WHY?
 - Build community
 - Connect with other students
 - Connect with Prof. Nicholson
 - See material from multiple perspectives by engaging in active learning
 - It's part of your overall grade! (**You MUST have an iClicker!**)
- Keep up with lectures and material --- biochemistry will be much more fun!
- Utilize the many resources provided for helping you learn the material
 - Join one of the uTA-led Study Groups
 - Attend the gTA Review Session and/or office hours
 - Attend Professor Nicholson's office hours

Assignments and Methods of Assessing Student Achievement

In-class participation

No makeups. Participation in **80%** of **in-class** activities earns full participation credit. In-class participation is important for collaborative learning and for keeping up with the material. We will engage in in-class activities that will include **iClicker** questions and small group problem-solving. **You MUST have a physical iClicker because an RF receiver will be used in the classroom.**

Weekly assignments/Homework – (working with others is encouraged!)

- Daily problems from the Lecture Guide (LG). **You are encouraged to work collaboratively, this is not graded!** See last part of LG and brief answers at back of LG. We will not collect or grade these problems, but *similar ones will appear on quizzes and exams.*
- Case Studies (online through course Canvas site). **Your best 3** of the 8 listed Case Studies are counted. **You are encouraged to work collaboratively! DUE by last day of class (Dec. 7).**
- ChimeraX assignments and Canvas ChimeraX Quizzes. **ChimeraX Tutorial** (in ChimeraX module) instructs you on how to download and install the ChimeraX software on your computer. For **most weeks**, a ChimeraX Canvas Quiz **will be due**. Use ChimeraX to work through each assignment and complete the corresponding **ChimeraX Canvas Quiz**. **You are encouraged to work collaboratively** on the assignments and the ChimeraX Quizzes! **Due dates** for each of the **10 ChimeraX Quizzes** are **listed in the Course Schedule above. ALL are counted, but 4 may be completed late (by the last day of class) to allow for illness, etc.**

Exams:

- **WEEKLY QUIZZES:** There will be a total of **10 weekly quizzes**
 - In-class Quiz held Wednesdays at the **BEGINNING OF CLASS, 15 minute** duration
 - Canvas Quiz version of same quiz will be posted after Wed. class; you can work collaboratively, open note, open book, and **submit BEFORE the next class (Friday).**
 - Each weekly quiz score = $0.8 \times \text{in-class score} + 0.2 \times \text{Canvas Quiz score}$
 - If Canvas Quiz is NOT SUBMITTED, weekly quiz score = $1.0 \times \text{in-class score}$
 - Each weekly quiz covers all not-yet-tested-on lectures through the preceding week (i.e., does NOT include the current week), typically the MWF lectures from the previous week but sometimes including 1-2 additional prior but not-yet-quizzed-on lectures.
 - **NO MAKEUPS, your best 6 Quiz scores** count toward overall Quiz grade (i.e., **4 drops**)
- **EVENING PRELIMS:** Two cumulative midterm prelims will be held at 7:30 – 9:00 pm:
 - Oct. 1 (on material covered 8/24 – 9/25)
 - Nov. 5 (on material covered 9/28 – 10/30).
 - If you require a make-up exam for a prelim, contact Prof. Nicholson.
- **FINAL EXAM:** A **fully cumulative** final exam will be held on a day and time during the final exam period that will be scheduled by the University.

Course Grading - Grade Components:

BioMG 3310

Assignment, Assessment or activity	Percentage of grade
In-Class Participation (iClicker Qs, Worksheets)	5%
In-Class Quizzes (best 6 out of 10 are counted)	20%
ChimeraX Quizzes (all are counted)	5%
Case Studies (best 3 out of all completed)	5%
Prelim 1 (7:30 – 9:00 PM, 10/1)	20%
Prelim 2 (7:30 – 9:00 pm, 11/5)	20%
Final Exam	25%

BioMG 6311 (Graduate Students Only)

Assignment, Assessment or activity	Percentage of grade
In-Class Participation (iClicker Qs, Worksheets)	2%
In-Class Quizzes (best 6 out of 10 are counted)	20%
ChimeraX Quizzes (all are counted)	1.5%
Case Studies (best 3 out of all completed)	1.5%
Prelim 1 (7:30 – 9:00 PM, 10/1)	20%
Prelim 2 (7:30 – 9:00 pm, 11/5)	20%
Commentary Article	10%
Final Exam	25%

Grading Scale: A criterion-based grading scale is employed; *your grade is independent of how anyone else performs in the class*. Please help each other learn!

LETTER	%	LETTER	%
A+	99– 100	C+	73 – 76.9
A	93 – 98.9	C	69 – 72.9
A-	90 – 92.9	C-	65 – 68.9
B+	86 – 89.9	D+	61 – 64.9
B	81 – 85.9	D	57 – 60.9
B-	77 – 80.9	D-	50 – 56.9
		F	< 50

Course Grading – Grading Practices and Policies:

- Re-grading requests for **In-Class quizzes**:
 - Graded Wednesday In-Class quizzes will be **posted in Gradescope** the next FRIDAY at the END OF CLASS.
 - **Submit regrade requests within Gradescope before the beginning of class on the next MONDAY.** In your request, **please specify**:
 1. WHAT you want to be considered for regrade
 2. YOUR RATIONALE for why you think your answer is correct
 - **No late regrade requests will be granted.**
- Re-grading requests for **Prelims**:
 - Regrade requests **must be submitted before the beginning of class one week after grades are posted**.
 - To request regrades for any **hand-written** problems, please **attach** a **cover sheet to your prelim exam paper** and specify:
 - WHAT you want to be considered for regrade
 - YOUR RATIONALE for why you think your answer is correct
 - To request regrades for any **Scantron** problems, **submit regrade requests within Gradescope**. In your request, please specify:
 - WHAT you want to be considered for regrade
 - YOUR RATIONALE for why you think your answer is correct
 - **No late regrade requests will be granted.**
- Keeping up with course content when absent:
 - It is each student's responsibility to keep up with missed course content when absent.
 - The 3310 Canvas site has all the material necessary to keep up with missed lectures.
 - Because 4 of the 10 in-class quizzes are dropped, **no missed quizzes can be made up**.
 - Similarly, because 20% of the in-class participation is dropped, there is **no make-up for missed participation points**.

Course Management and Policies

Academic Integrity:

I expect absolute integrity of every Cornell student in all academic work. This means that I expect every student in my class to follow Cornell's Code of Academic Integrity. This Code is based on a set of values. These values are grounded in honesty in your own intellectual efforts, and honesty with respect to the work of your classmates. I expect academic integrity in all formal coursework, AND in all University relationships and interactions connected to the academic process, for example, in the use of University resources. **Any work submitted by a student in this course for academic credit must be the student's own work or credit will not be given.**

You are responsible for knowing and understanding Cornell's policies regarding academic integrity, which are available here: <https://www.deanoffaculty.cornell.edu/faculty-and-academic-affairs/academic-integrity/code-of-academic-integrity/>

In addition to the more familiar aspects of the code, ***here are a few highlighted specifics that might be less familiar regarding Academic Integrity expectations in BIOMG 3310/6311:***

Students agree that by taking this course, each student **will have in their possession ONLY THEIR OWN iCLICKER, registered to them.** If a student is observed with more than one iClicker in class, all iClickers in their possession will be taken by Professor Nicholson or a TA for identification purposes, and **all involved students** (the student in possession and all students registered to the taken iClickers) will be charged with an **Academic Integrity Violation.**

Students agree that by taking this course, they will **not record (video or voice) in the BIOMG 3310 classroom** without **prior written approval from the instructor.** See Sections I.A.4, I.B.4 and I.C.3 in <https://www.deanoffaculty.cornell.edu/faculty-and-academic-affairs/academic-integrity/code-of-academic-integrity/> . **If a student is observed holding a phone or other recording device in a manner that suggests they are recording in the classroom,** Professor Nicholson will stop class and ask them to put their phone or device away. If they do not comply, they will be asked to leave the classroom. If they need to be asked a second time, they will be charged with an **Academic Integrity Violation.**

For BIOMG 6311 students: Students agree that by taking this course, the required Commentary paper may be subject to submission for textual similarity review to Turnitin.com for the detection of **plagiarism, and GAI use is not permitted.** All submitted papers will be included as source documents in the Turnitin.com reference database solely for the purpose of detecting plagiarism of such papers. Use of the Turnitin.com service is subject to the Usage Policy posted on the Turnitin.com site.

SDS Accommodations

Students with Disabilities: Your access in this course is very important to me. In order to have adequate time to arrange your approved accommodation(s), you must request your [SDS accommodation letter](#) no later than the add/drop deadline for the semester.

- Students currently registered with SDS: Once you request your accommodation letter and it is approved by SDS, it will be emailed to both you and me. Processing time can be up to 48-hours.
- Students not registered with SDS: The registration process for new accommodations can take up to three weeks. Once you are approved by SDS for accommodations, you will be able to request your accommodation letter for this course.
- If you are approved for accommodations later in the semester: you must request your accommodation letter as soon as possible.

Students with Exam Accommodations: In addition to requesting your accommodation letters, this course will be participating in the Alternative Testing Program (ATP). All accommodated exams will be centrally managed and supported by the ATP Testing Coordinator in the Office of Student Disability Services and accessible in your SDS student portal. ATP support includes:

- Scheduling of accommodated exams:
 - **In-Class Quizzes:** Students are expected to **join the class at 10:25 am at the end of the general in-class quiz administered during class time.**
 - Students may select to take their accommodated quiz at 8 a.m. and/or 5 p.m. on the **same day** as the general quiz. More information is available here at sds.cornell.edu/atp
 - **Evening Prelims:** The evening prelim exams will begin promptly at 6:30 p.m for SDS-accommodated students, or as otherwise scheduled with you by SDS's ATP service.
- Notification of accommodated quiz/exam logistics:
 - All quiz/exam logistics are managed by the ATP and will ONLY be communicated to you via email from sds-testing@cornell.edu and accessible in your SDS student portal. Please do not contact me with questions about exam logistics, as I will not be able to answer them.
 - **10-days prior to the exam date:** ATP will automatically send an email with the exam date, time, location.
 - **48-hours prior to the exam date:** ATP will send a reminder email about the exam.
- Coordination of **make-up** Prelim or Final exams (i.e., for students who have been granted prior permission by me to take the exam on a day other than the scheduled date of the main exam) will be handled by me (Professor Nicholson). The ATP will not be involved in the logistics for any make-up exams. If you miss your scheduled accommodated exam, you should notify **Professor Nicholson**, not the ATP.

For students with other academic accommodations (not testing-related), please follow up with Professor Nicholson to discuss the necessary logistics of your accommodation(s).

Mental Health and Stress Management Resources

If you are experiencing undue personal or academic stress at any time during the semester or need to talk with someone about a personal matter or situation, I encourage you to seek support immediately. I (Professor Nicholson) am available to talk with you about stresses related to your work in my class.

Additionally, I can assist you in reaching out to a wide range of campus resources, including:

- Your college's Academic Advising or Student Services Office
- Cornell Learning Strategies Center at 607-255-6310, learningstrategiescenter@cornell.edu
- Cornell Health at 607-255-5155, <https://health.cornell.edu/about/contact-us>
- Let's Talk = Drop-in counseling, <https://health.cornell.edu/services/counseling-psychiatry/lets-talk>

Peer Mentoring at EARS = Empathy Assistance and Referral Service:

<https://www.earscornell.org> (213 Willard Straight Hall, 607-255-3277)

- Student Disability Services at 607-254-4545, sds_cu@cornell.edu
- If you experience illness that impacts your ability to fully participate in this class, please 1) request needed accommodations from SDS, 2) email Prof. Nicholson (lkn2), and 3) contact Student Services in your college/major.

Academic Concern Notice

At Cornell, we use an academic support practice called "Academic Concern" that's designed for all students, across every college in the university. It serves as a communication bridge among you, your instructors, and your [college academic advising office](#) that prompts outreach to support you when, for example, an exam didn't go as planned or if you've missed several classes (especially when those absences can affect your grade).

Here's how it works: If one of your instructors notices you might need academic support, they let your college advising office know. The office will then invite you to [schedule a meeting](#) with an advisor so you can be proactive in making a plan to keep your academics on track. They'll also point you to some valuable campus resources that can help you improve your learning.

Keep in mind, if we submit an Academic Concern, it's not because you're in trouble or have done something wrong. ***It's simply a way for us to make sure you're getting the support you need, when you need it, to be successful in this course.***